

Mining Social Networks

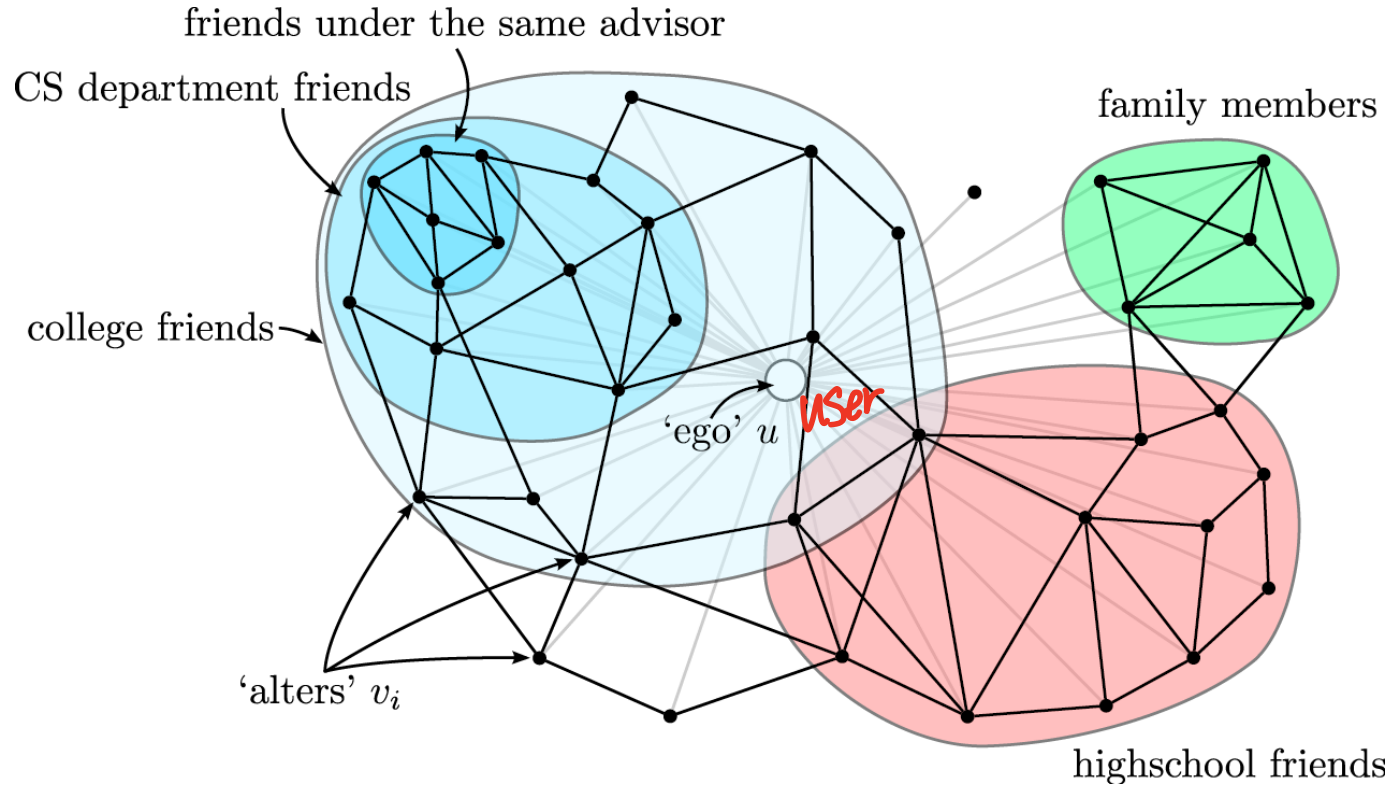
COMPSCI 753

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Overview

- Social network analysis
 - Social network introduction and properties
 - Community detection
 - Influence maximization

Online Social Networks (OSN)



[McAuley, Leskovec: Discovering social circles in ego networks, 2012]

Terminologies 术语.

- Given an undirected graph $G(V, E)$, define the following terms:
 - Neighbor $N(v)$** : the set of nodes connect to v - $N(v) = \{u | (u, v) \in E\}$
 - Degree d_v** of a node v : the number of neighbors of v
 - Distance $dist(s, t)$** : the shortest path length from s to t
 - Eccentricity** of a node $v \in V$: $ecc(v) = \max_{u \in V} \text{dist}(v, u)$
离心度
 - Radius r** : the minimum eccentricity, $\min_{v \in V} ecc(v)$
半径
 - Diameter d** : the maximum eccentricity, $\max_{v \in V} ecc(v)$
直径

OSN Properties – Small diameter

- Real-world networks tend to have small diameter

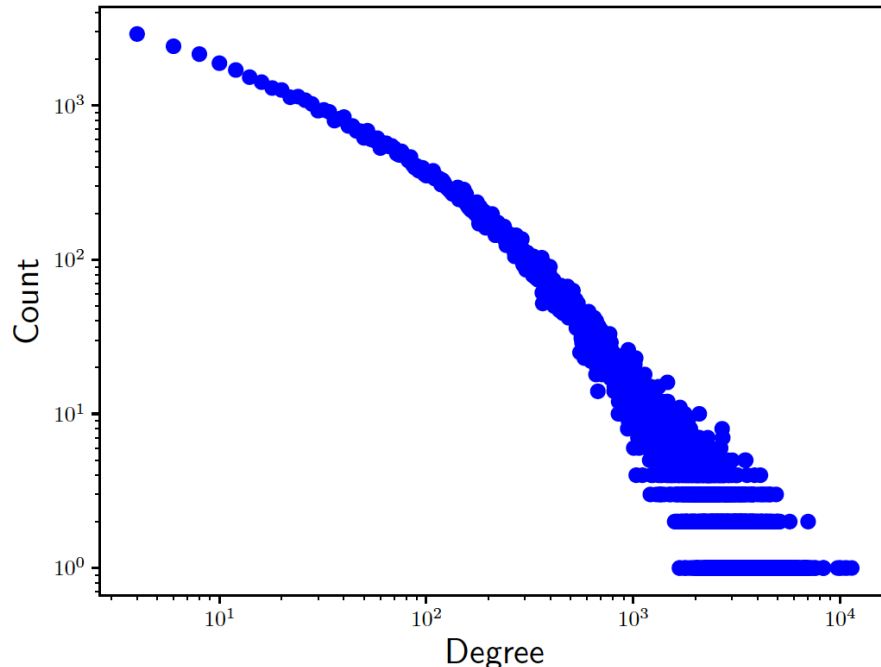
Statistical Information				
Dataset	n	m	r	d
Facebook	4 039	88 234	4	8
Brightkite	56 739	212 945	9	18
Epinions	75 877	405 739	8	15
Slashdot	77 360	469 180	6	12
Twitter	81 360	1 342 296	4	7
Gowalla	196 591	950 327	8	16
DBLP	317 080	1 049 866	12	23
Youtube	1 134 890	2 987 624	12	24
Flicker	1 624 992	15 476 835	12	24
Grqc	4 158	13 422	9	17
Hepth	8 638	24 806	10	18
Hepph	11 204	117 619	7	13
Astroph	17 903	196 972	8	14
Condmatt	21 363	91 286	8	15
Askubuntu	152 599	453 221	7	13
Superuser	189 191	712 870	6	12
Wiki-vote	7 066	100 736	4	7
Wiki-temporal	1 091 742	2 786 764	5	9
Wiki-talk	2 388 953	4 656 682	6	11
Web-stanford	266 388	2 228 348	5	9

Six Degrees of Separation

There is a famous game called “six degrees of Kevin Bacon,” the object of which is to find paths of length at most six in the graph whose nodes are movie stars and whose edges connect stars that played in the same movie. The conjecture is that in this graph, no movie star is of distance more than six from Kevin Bacon. More generally, any two movie stars can be connected by a path of length at most six; i.e., the diameter of the graph is six. A small diameter makes computation of neighborhoods more efficient, so it would be nice if all social-network graphs exhibited a similar small diameter. In fact, the phrase “six degrees of separation,” refers to the conjecture that in the network of all people in the world, where an edge means that the two people know each other, the diameter is six. Unfortunately, as we shall discuss in Section 10.8.3, not all important graphs exhibit such tight connections.

OSN Properties - Power law degree distribution

- Minority of the nodes have large degree
- Most have few neighbors



Degree distribution on Flickr (from
<http://konect.cc/networks/flickr-growth/>)

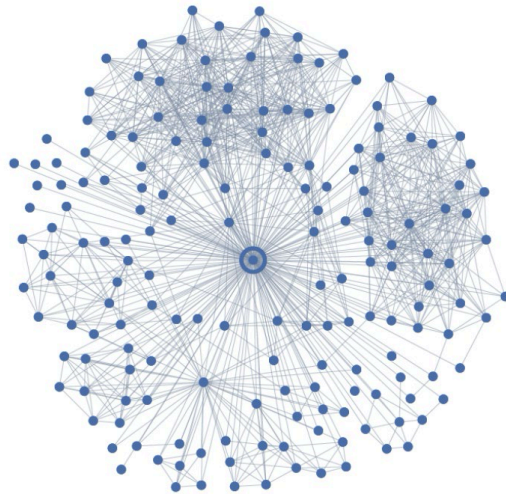
Core-Periphery Structure: A social network has a densely connected **core**.

Intuition: The rich gets richer, winners take all, small advantages are magnified to a critical mass, new ideas that get attention become “viral”.

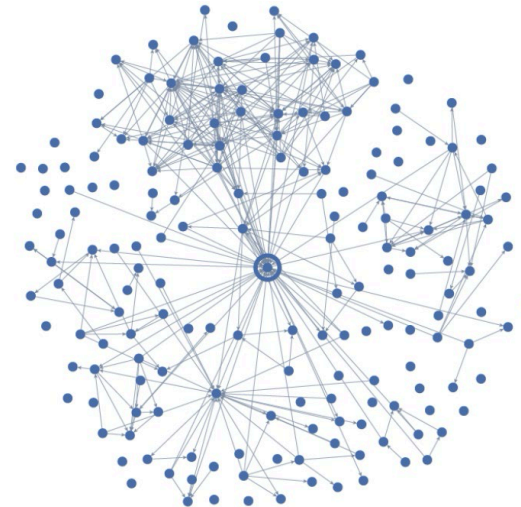
OSN Properties – Triadic closure

- **Strong ties:** Close and frequent social contact
- **Weak ties:** More casual and distinct social contact

Triadic Closure: Two friends of a user tend to become friend in the future



All friends



Close friends

Clustering Coefficient 聚类系数

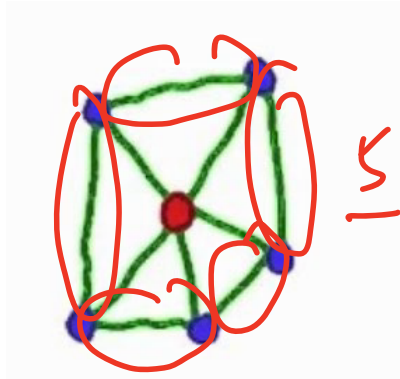
- How likely the friends of a node is are also friends?
- We can use a measure the degree to which nodes in a graph tend to cluster together
- Clustering coefficient: the local clustering coefficient $CC(u)$ of a single node u measures how strong u connects to others:

$$CC(u) = \frac{\text{\# of edges between the neighbors of } u}{\text{\# of node pairs in the neighbors of } u} = \frac{|\{(v,w) \in E\} : v,w \in N(u)|}{\binom{|N(u)|}{2}}$$

where $N(u)$ - Neighbors of u

$N u (N u - 1) / 2$

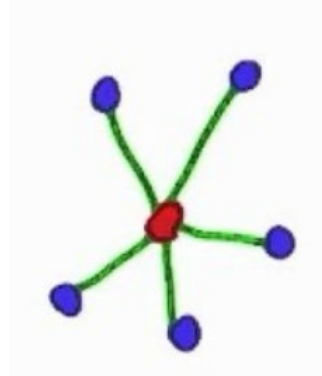
Example



$$N(u) = 5$$

$$CC(u) = \underline{5}/10 = 1/2$$

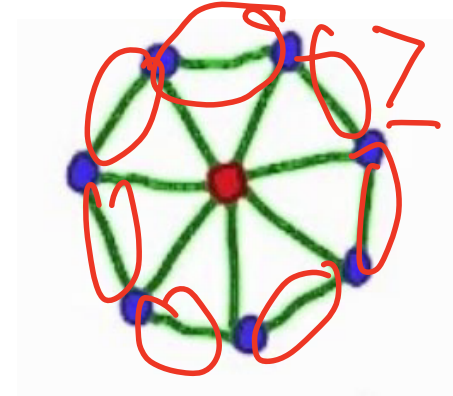
$$\frac{4 \times 5}{2} = 10$$



$$N(u) = 5$$

$$CC(u) = 0/10 = 0$$

No tendency to cluster at all!



$$N(u) = 7$$

$$CC(u) = \underline{7}/21 = 1/3$$

$$\frac{7 \times 6}{2} = 21$$

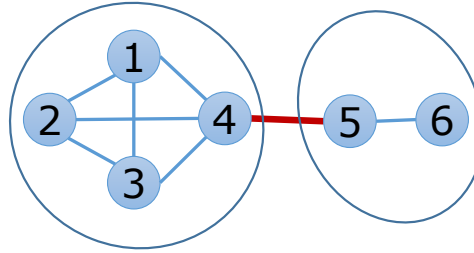
More neighbours, but not necessarily higher tendency to cluster

Overview

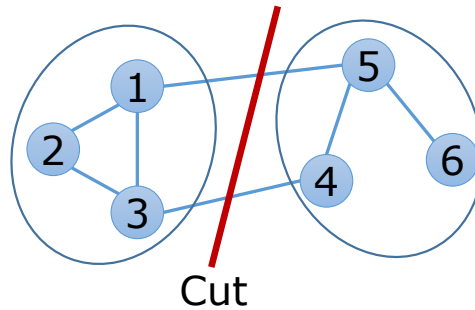
- Social network analysis
 - Social network introduction
 - **Community detection**
 - Influence maximization

How to define good communities?

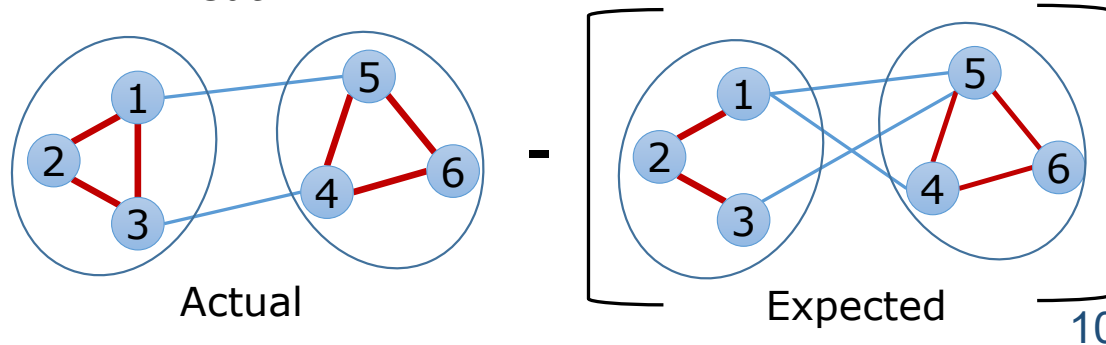
- Edge betweenness



- Best cuts

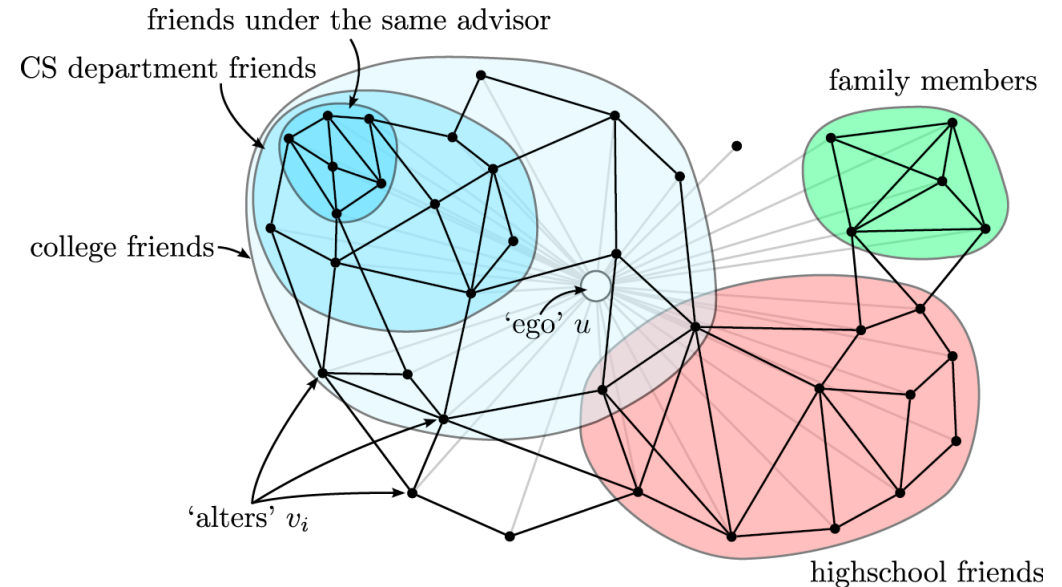


- Modularity



Community detection

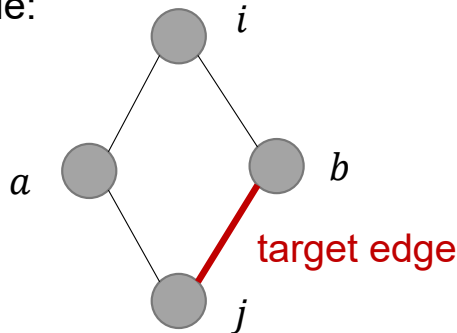
- **Brandes and Girvan-Newman algorithms**
- Spectral clustering
- Modularity maximization



Computing edge betweenness

- How can we measure the **importance of an edge** as a bridge between communities?
- Edge betweenness:** Number of shortest paths from one node to another passing through the edge
 - Each pair of source/destination nodes contributes **at most 1 shortest path** to the edge betweenness measure.
 - If N shortest paths exist between the two nodes and only k of them pass through the edge, then k/N shortest paths will contribute to the edge's betweenness.

Example:



Two shortest paths from i to j :

- $i \rightarrow a \rightarrow j$
- $i \rightarrow b \rightarrow j$

Only $i \rightarrow b \rightarrow j$ pass through the target edge

The shortest path $i \rightarrow b \rightarrow j$ **contributes 0.5** to the edge betweenness of the target edge!

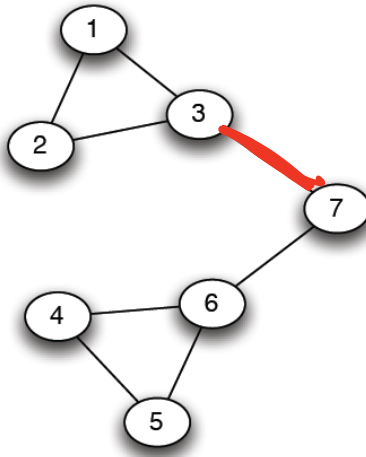
$$\frac{1}{2}$$

targeted edge.

Computing edge betweenness 中心度

- How can we measure the importance of an edge as a bridge between communities?
- Edge betweenness:** Number of shortest paths from one node to another passing through the edge

Example:



Edge betweenness of edge (3,7)?

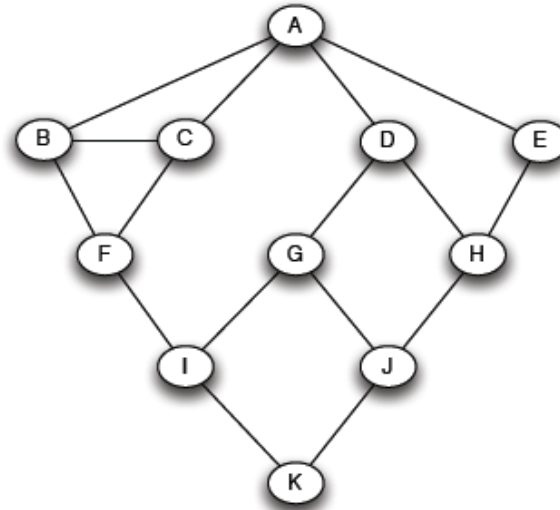
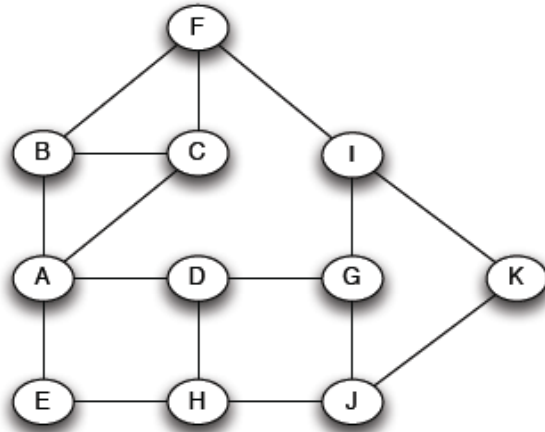
6x6

- Shortest path from **node 1** to **node 2**, doesn't pass through (3,7)
- Shortest path from **node 1** to **node 3**, doesn't pass through (3,7)
- Shortest path from **node 1** to **node 4**, pass through (3,7)
- Shortest path from **node 1** to **node 5**, pass through (3,7)
- Shortest path from **node 1** to **node 6**, pass through (3,7)
- Shortest path from **node 1** to **node 7**, pass through (3,7)
- Shortest path from **node 2** to **node 1**, doesn't pass through (3,7)

.....

Brandes' algorithm

- Ulrik Brandes, 2001
- Idea on **unweighted** graph – breadth-first-search, level = distance



level 0

level 1

level 2

level 3

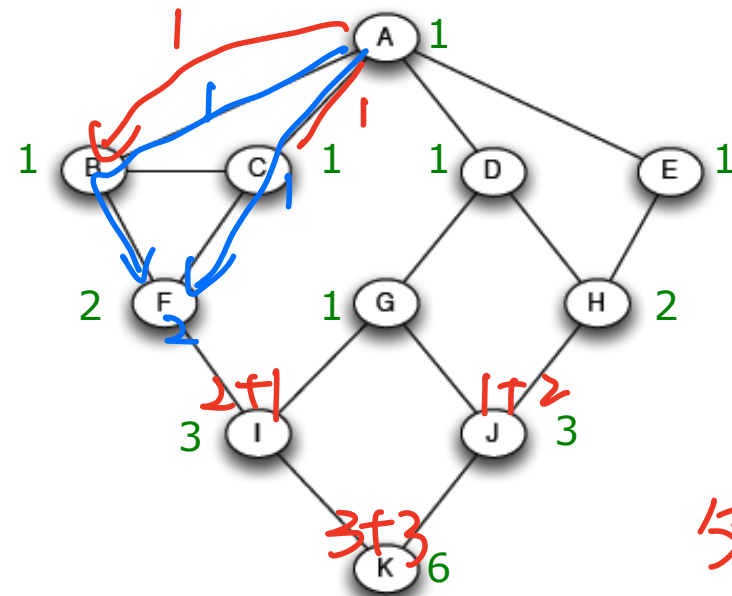
level 4

Brandes' algorithm

- Step 1: **Top-down** to compute # shortest path from root to a node
- Step 2: **Bottom-up** to distribute # shortest path to edges

Top-down:

- Start from root and assign 1 as #shortest path to root (level 0)
- Nodes on level 1 will also be assigned 1 as #shortest path
- Starting on level 2:
For each node, add #shortest path of its parents



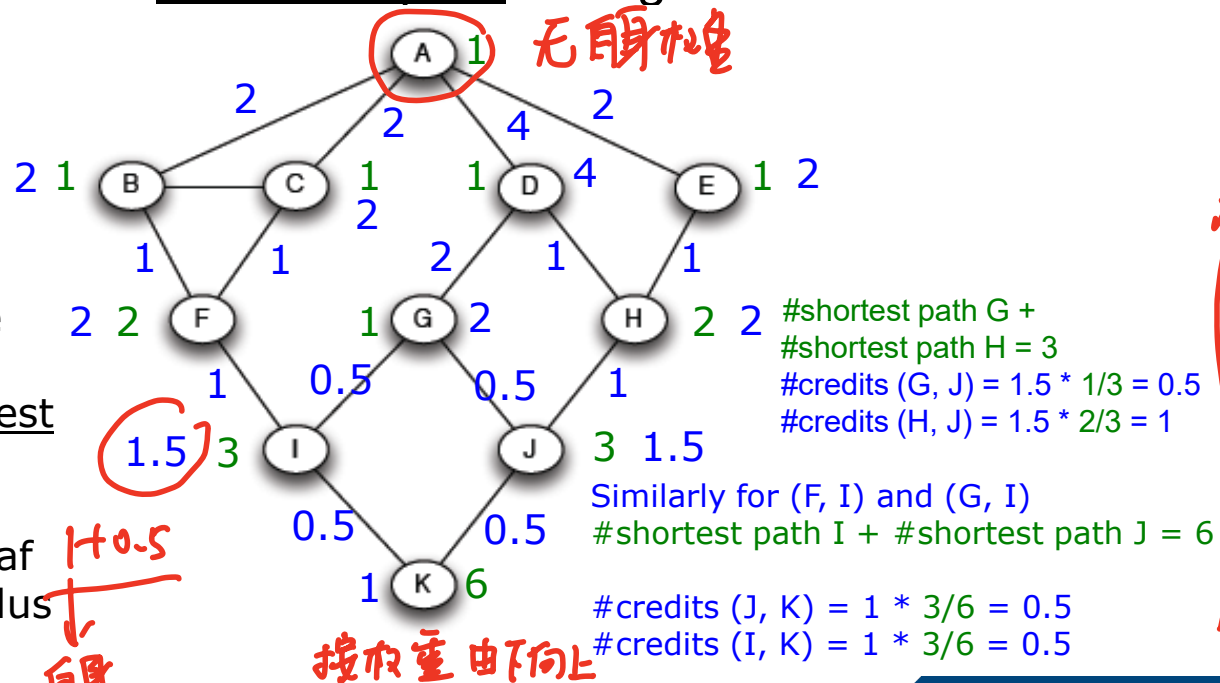
父节点的集合

Brandes' algorithm

- Step 1: **Top-down** to compute # shortest path from root to a node
- Step 2: **Bottom-up** to distribute # shortest path to edges

Bottom-up:

- Each leaf node gets 1 credit
- Start from leaves, then backpropagate the credits to both edges and nodes:
 - Distribute the credit of the node to each **edge** proportional to the #shortest path of the node's parents (step 1)
 - Each **node** that is not a leaf gets a credit equals to 1 plus the sum of credits of the edges from that node to the level below



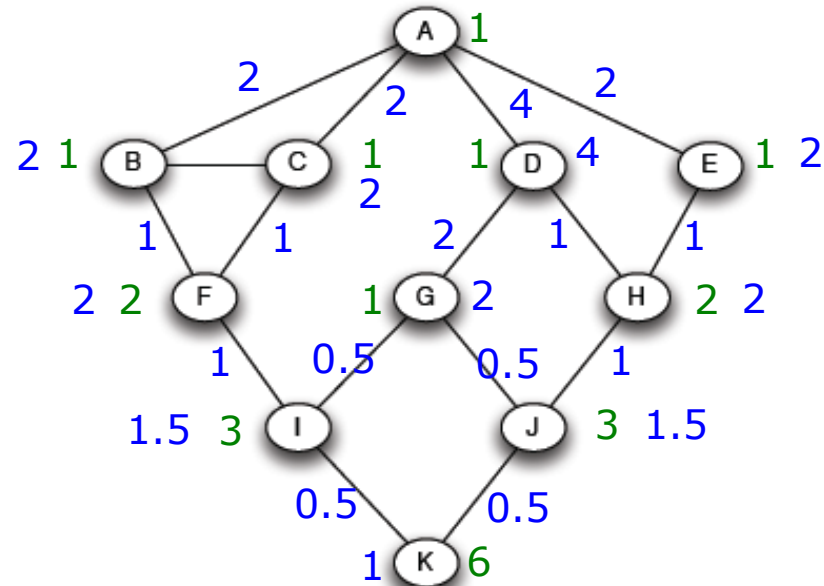
Brandes' algorithm

- Step 1: **Top-down** to compute # shortest path from root to a node
- Step 2: **Bottom-up** to distribute # shortest path to edges

Complexity

- We need to traverse the (unweighted graph) with BFS from top, and the reverse from bottom
- BFS takes $O(|E|)$ time
- Need to run BFS for every node!

Overall complexity – $O(|E||V|)$ go through,
all edges ↙ all nodes ↘

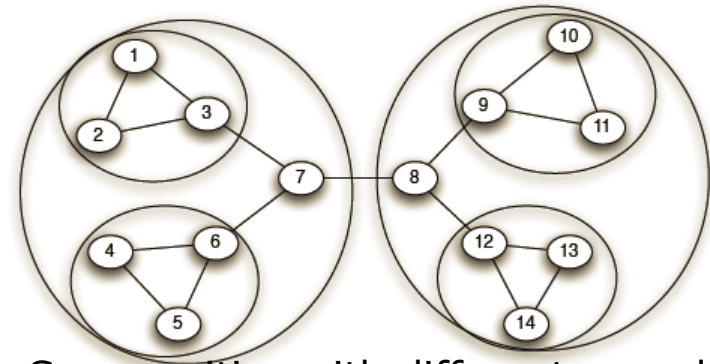
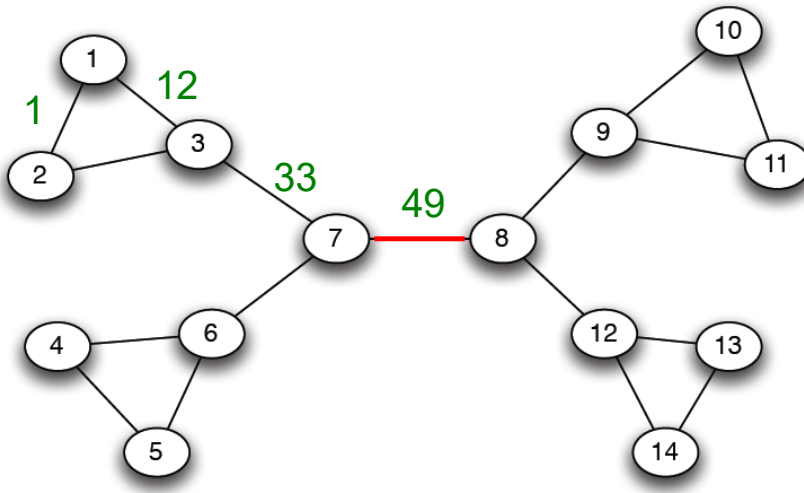


Girvan-Newman algorithm

- Divisive hierarchical clustering based on **edge betweenness**

- **How?**

1. Apply Brandes' algorithm to compute betweenness, for every node as the root, and sum the contributions. Divide this by 2, since every shortest path will be discovered twice, one for each of its endpoints!
2. Remove the edge(s) with highest betweenness
3. Repeat 1 and 2 until no further edges are left or we get a number of k clusters



Communities with different granularities